

On evolutionary stability of carrying capacity driven dispersal in competition with regularly diffusing populations

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Abstract

Two competing populations in spatially heterogeneous but temporarily constant environment are investigated: one is subject to regular movements to lower density areas (random diffusion) while the dispersal of the other is in the direction of the highest per capita available resources (carrying capacity driven diffusion). The growth of both species is subject to the same general growth law which involves Gilpin-Ayala, Gompertz and some other equations as particular cases. The growth rate, carrying capacity and dispersal rate are the same for both population types, the only difference is the dispersal strategy. The main result of the paper is that the two species cannot coexist (unless the environment is spatially homogeneous), and the carrying capacity driven diffusion strategy is evolutionarily stable in the sense that the species adopting this strategy cannot be invaded by randomly diffusing population. Moreover, once the invasive species inhabits some open nonempty domain, it would spread over any available area bringing the native species diffusing randomly to extinction. One of the important technical results used in the proofs can be interpreted in the form that the limit solution of the equation with a regular diffusion leads to lower total population fitness than the ideal free distribution.

Keywords: ideal free distribution; evolutionary stability; Gilpin-Ayala equation, Gompertz growth rate, carrying capacity driven diffusion, global attractivity, system of partial differential equations, competition, semi-trivial equilibria

AMS subject classification: 92D25, 35K57 (primary), 35K50, 37N25

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1 Introduction

Incorporation of spatial distribution in equations of population dynamics is a very important step in adequate modeling of real world phenomena. The only method to describe species invasion, stocking and pattern formation is to include movements in equations. The simplest way to introduce dispersion mimics particle diffusion in homogeneous environments assuming the flow from higher to lower density. Recently there were several attempts to involve alternative types of movements, for example, slower speed increase with the growth of density gradients by Wang et al. (2013), see also references therein. However, in heterogeneous environments the main problem is that the diffusion types adopted from physical models assume massive migration of individuals to the areas of less per capita available resources, where they are destined to extinction, as long as these areas have lower population density. Expanding this idea to human population models would contradict to the existence of large urban areas with high population density and high availability of economic and social resources. Whatever type of regular diffusion is accepted, the population would tend to be uniformly distributed over space as the diffusion coefficient grows, while the ideal free distribution would correspond to equal availability of per capita resources at each location. This model has a physical analogue of particle distribution in an inhomogeneous field where stronger particle repulsion corresponds to the areas of lower carrying capacity values. The ideal free distribution describes how species can distribute themselves to maximize their total fitness, as noted by Fretwell and Lucas (1970); any movement in an ideally distributed population will lead to decrease in fitness. However, random diffusion models, generally, cannot lead to this type of dispersal in spatially heterogeneous environments.

As a remedy to this situation, the random diffusion was combined with the directed movement along the environmental gradient, see Cantrell et al. (2007), Cantrell, Cosner and Lou (2007), Cantrell et al. (2008), Cantrell et al. (2010) and also the recent papers by Cantrell et al. (2012), Chen et al. (2012) and Zhang et al. (2012). This dispersal type may corresponds to the reaction-diffusion-advection model introduced by Cantrell et al. (2008). For such equations, solutions tend to be ideally distributed, if the ratio of the advection to the dispersal coefficients tends to infinity. In reaction-diffusion-advection models considered by Cantrell, Cosner and Lou (2007) and Chen et al. (2012) there is advection but because of its form it cannot lead to an ideal free distribution no matter how strong it is relative to diffusion. An ideal free distribution can be achieved for a specific finite rate of advection in the model considered by Cantrell et al. (2010), and this result was recently improved by Averill et al. (2012).

The dispersal type which has the ideal free distribution as a solution was introduced by Braverman and Braverman (2009) as a part of an optimal harvesting problem. It was demonstrated by Korobenko and Braverman (2009) and Korobenko et al. (2013) that for a wide class of equations and time-constant carrying capacities all positive solutions tend to the ideal free distribution. Further, this type of dispersal will be referred to as a carrying capacity driven diffusion. The case of spatially heterogeneous but temporally constant environment is very important and was studied in many papers, see Cantrell et al. (2007),

Cantrell, Cosner and Lou (2007), Cantrell et al. (2008), Cantrell et al. (2010), Cantrell et al. (2012), Chen et al. (2012). If the carrying capacity is time-dependent, positive solutions of the equation with a carrying capacity driven diffusion, generally, do not tend to the ideal free distribution, as Example 3 of the present paper demonstrates, but the strategy leading to this distribution is still advantageous.

One of the important characteristics of adaptive dynamics is evolutionary stability, see the paper by Dieckmann and Law (1996) for historical reference and the monograph by Geritz and Gyllenberg (2008) for the detailed outline. The evolutionary stability of strategies of the type discussed by Cantrell et al. (2008) was recently studied in Cantrell et al. (2013).

An evolutionarily stable strategy adopted by a resident population makes it impossible for a population choosing a different strategy to invade its habitat. Moreover, as we demonstrate in the present paper, carrying capacity driven dispersal strategy allows a population to invade any available area, if the density of invading species is positive on an arbitrarily small original habitat. However, evolutionary stability in a global sense is not always easy to analyze compared to local characteristics described by convergent stable strategies. A strategy is considered convergent stable if small deviations from this strategy lead to disadvantage. For example, it was demonstrated by Cantrell et al. (2010) that advection in the direction of maximal available resources is a convergent stable strategy. The same approach was applied by Dockery et al. (1998) to deduce that among several randomly diffusing populations which differ by the diffusion coefficient only, the slowest population has an evolutionary advantage. Whenever both random dispersal and advection are involved, the interplay of these two coefficients in each of two competing species was recently investigated by Chen et al. (2012). However, the results of the present paper are different from the following point of view: diffusion coefficients for the two competing populations can be any positive numbers. For any relation of the two, the population adopting carrying capacity driven dispersal will invade the habitat; lower diffusion rates only slow down this process, with the same eventual result.

The idea that a null model for habitat patch selection is the ideal free distribution was outlined by Schreiber (2012). The tendency of populations to disperse according to the inhomogeneous carrying capacity is observed for grazing animals by Oosterheld and Semmartin (2011). For *Solanum carolinense* brought from North America to Europe, without control the invasive species will spread over the whole continent, according to the available resources, as shown by Follak and Strauss (2010); mainly, climate-related issues were considered. As illustrated by Padron and Trevisan (2006), for nonhomogeneous environment purely diffusive behavior produces an unbalanced distribution when some locations receive more individuals than can be supported by the environmental carrying capacity, while others receive less. An evolutionarily stable strategy assumes an asymmetric migration mechanism, induced by the heterogeneous carrying capacity of the environment, as mentioned by Padron and Trevisan (2006). The environmental properties can include food supply productivity for herbivores, climate parameters, or human hosts for insects *Aedes aegypti*, as mentioned by Maciel-de-Freitas et al. (2010). The main result of the present paper is evolutionary stability of carrying capacity driven dispersal. The com-

petition model considered has only three stationary states: the zero and two semi-trivial equilibria obtained as a result of the competitive exclusion. The proof of the fact that there is no coexistence equilibrium in nonhomogeneous environment is based on Lemma 2 which can be interpreted in the way that the total fitness of the population adopting the random diffusion strategy is lower than of the one attained for the ideal free distribution.

The paper is organized as follows. Section 2 describes the competition model and verifies that it is properly posed: states existence and uniqueness results, as well as justifies positivity of all solutions with nonnegative (and not identically equal to zero) initial conditions. Section 3 includes the main results of the paper: there is no coexistence equilibrium, and the strategy leading to the ideal free distribution has the advantage of evolutionary stability. In Section 4 we consider a competition of two populations, one driven by the carrying capacity and another by another function, different from the carrying capacity. Section 5 demonstrates that the results are applicable to the variety of growth laws: logistic, Gilpin-Ayala, Gompertz and some others. It includes examples illustrating both higher and lower average population levels under the carrying capacity driven dispersal compared to the regular diffusion. The third example illustrates that once the environment is time-variable, it is possible to imagine a situation when the random, not carrying capacity driven dispersal leads to the ideal free distribution. Finally, Section 6 presents summary and discussion, and outlines some possible extensions of the model considered in the present paper.

2 Model Formulation and Positivity of Solutions

2.1 The model

We consider the following system of parabolic equations with the Neumann boundary conditions:

$$\begin{cases} \frac{\partial u(t, x)}{\partial t} = D\Delta \left(\frac{u(t, x)}{K(x)} \right) + u(t, x)g(x, u(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial v(t, x)}{\partial t} = \nabla \cdot d(x)\nabla v(t, x) + v(t, x)g(x, v(t, x), u(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial(u/K)}{\partial n} = d(x)\frac{\partial v}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (2.1)$$

with the initial conditions

$$u(0, x) = u_0(x), \quad v(0, x) = v_0(x). \quad (2.2)$$

We assume that $D > 0$, $K(x)$ is in the class $C^{1+\alpha}(\overline{\Omega})$, $K(x) > 0$ for any $x \in \overline{\Omega}$. For the diffusion coefficient of the second species with a regular diffusion, we assume $d(x) > 0$ in $\overline{\Omega}$ and $d(x) \in C^1(\overline{\Omega})$. Here Ω is a an open nonempty bounded domain of $\mathbb{R}^n$ with $\partial\Omega \in C^{2+\alpha}$, $0 < \alpha < 1$, and $J_1 \times J_2$ a bounded subset of $\mathbb{R}^2$. The set $J_1 \times J_2$ corresponds to the range of the solutions to (2.1) and is determined by the corresponding upper and

lower solutions. It is also convenient to introduce the following notation

$$Q = (0, \infty) \times \Omega, \quad \overline{Q} = [0, \infty) \times \overline{\Omega}, \quad \partial Q = (0, \infty) \times \partial\Omega$$

To state the assumptions on the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$, we will need the following definition (Pao 1992).

Definition 1. A function $f_i = f_i(u_1, u_2)$ is said to be quasimonotone nonincreasing if for fixed u_i , f_i is nonincreasing in u_j for $j \neq i$. A vector-function $\mathbf{f} = (f_1, f_2)$ is called quasimonotone nonincreasing in $J_1 \times J_2$ if both f_1 and f_2 are quasimonotone nonincreasing for $(u_1, u_2) \in J_1 \times J_2$.

We next list some assumptions on the growth functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ which will be used throughout the paper:

- (A1) $f_i(\cdot, u, v)$, $i = 1, 2$ is Hölder continuous in Ω and $f_i(x, \cdot, \cdot) \in C^2(J_1 \times J_2)$;
- (A2) $\mathbf{f} = (f_1, f_2)$ is quasimonotone nonincreasing in $J_1 \times J_2$;
- (A3) $g(x, u_1, u_2) = g(x, u_2, u_1) = G(x, u_1 + u_2)$;
- (A4) $G(x, K) = 0$;
- (A5) $G(x, u)$ is strictly monotonically decreasing in u , $G(x, u) < 0$ when $u > K$ and $G(x, u) > 0$ when $0 < u < K$, $(t, x) \in Q$.

There are numerous examples of functions $G(x, u)$ satisfying the above properties, some of them are given in Section 5.1. However, the property (A5) is not satisfied for growth functions with the Allee effect described by Kot (2001).

2.2 Existence, uniqueness and positivity of solutions

Consider the initial-boundary value problem with the Neumann boundary condition and the carrying capacity driven diffusion

$$\frac{\partial u(t, x)}{\partial t} = D\Delta \left(\frac{u(t, x)}{K(t, x)} \right) + u(t, x)G(x, u(t, x)), \quad (t, x) \in Q \quad (2.3)$$

$$\frac{\partial \left(\frac{u}{K} \right)}{\partial n} = 0, \quad (t, x) \in \partial Q \quad (2.4)$$

$$u(0, x) = u_0(x), \quad x \in \Omega, \quad (2.5)$$

First, we establish some results for scalar equations of the form (2.3)-(2.5) which will be used in our further investigation of systems. The next two theorems were obtained by Korobenko et al. (2013).

Theorem 1. Let $f(x, u) = uG(x, u)$ satisfy properties (A1), (A4), and (A5) and let $u_0(x) \in C(\Omega)$, $u_0(x) \geq 0$ in Ω and $u_0(x) > 0$ in some open bounded nonempty domain $\Omega_1 \subset \Omega$. Then there exists a unique solution $u(t, x)$ of the problem (2.3)-(2.5) and it is positive.

Theorem 2. Let $f(x, u) = uG(x, u)$ satisfy properties (A1), (A4), and (A5). Then the function $u(x) := K(x)$ is an equilibrium solution of (2.3)-(2.5). Moreover, for any $u_0(x) \geq 0$, $u_0(x) \not\equiv 0$ the solution $u(t, x)$ of (2.3)-(2.5) satisfies

$$\lim_{t \rightarrow \infty} u(t, x) = K(x)$$

uniformly in $x \in \overline{\Omega}$.

Proof. Note that all the conditions of Theorem 6 obtained by Korobenko et al. (2013) are satisfied, therefore there exists a unique positive equilibrium solution u^* of (2.3)-(2.5). It is easy to check that $u^* = K$. $\square$

We will also need existence and convergence results for scalar systems involving a classical diffusion term

$$\begin{aligned} \frac{\partial v(t, x)}{\partial t} &= \nabla \cdot D(x) \nabla v(t, x) + v(t, x) G(x, v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial v}{\partial n} &= 0, & t > 0, x \in \partial\Omega, \\ v(0, x) &= v_0(x), & x \in \Omega \end{aligned} \quad (2.6)$$

The next result coincides with Theorem 1.14 in the monograph by Cantrell and Cosner (2003).

Theorem 3. Let $f(x, u) = uG(x, u)$ satisfy properties (A1), (A4), and (A5) and let $v_0(x) \in C(\Omega)$, $v_0(x) \geq 0$ in Ω and $v_0(x) > 0$ in some open bounded nonempty domain $\Omega_1 \subset \Omega$. Then there exists a unique solution $v(t, x)$ of the problem (2.6) and it is positive.

Theorem 4. Let $f(x, u) = uG(x, u)$ satisfy properties (A1), (A4), and (A5). Then there exists a unique positive equilibrium $v^*(x)$ of the problem (2.6). Moreover, for any $v_0(x) \geq 0$, $v_0(x) \not\equiv 0$ the solution $v(t, x)$ of (2.6) satisfies

$$\lim_{t \rightarrow \infty} v(t, x) = v^*(x)$$

uniformly in $x \in \overline{\Omega}$.

Proof. It is easy to check that all the conditions of Propositions 3.2 and 3.3 in the monograph by Cantrell and Cosner (2003) are satisfied and therefore the conclusion holds, see also Proposition 4 in Appendix of the paper by Korobenko et al. (2013). $\square$

The next theorem provides existence and uniqueness of solutions for coupled systems of equations.

Theorem 5. *Let the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1), (A2), and, in addition, $g(x, u, v)$ be strictly decreasing in u and v . Then for any $u_0(x), v_0(x) \in C(\Omega)$, problem (2.1) has a unique solution (u, v) . Moreover, if the initial condition $(u_0(x), v_0(x))$ is nonnegative and nontrivial, then $u(t, x) > 0$, $v(t, x) > 0$ for any $t > 0$.*

Proof. We will apply Theorem 16 from Appendix to the system obtained after the substitution $w(t, x) = u(t, x)/K(x)$ in (2.1)

$$\begin{cases} \frac{\partial w(t, x)}{\partial t} = \frac{D}{K(x)} \Delta w(t, x) + w(t, x) \tilde{g}(x, w(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial v(t, x)}{\partial t} = \nabla \cdot d(x) \nabla v(t, x) + v(t, x) \tilde{g}(x, w(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial w}{\partial n} = d(x) \frac{\partial v}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \quad (2.7)$$

where $\tilde{g}(x, w(t, x), v(t, x)) = g(x, K(x)u(t, x), v(t, x))$. Next, choose a constant ρ_w such that

$$\rho_w \geq \sup_{(t, x) \in Q_T} \frac{u_0(t, x)}{K(t, x)}$$

and

$$\tilde{g}(t, x, \rho_w, 0) < 0$$

which can be found since g is strictly decreasing in u . Note that since u_0 is bounded in $\overline{\Omega}$ and K is bounded from below, we have $\sup_{x \in \Omega} (u_0(x)/K(x)) < \infty$. Similarly, choose $\rho_v > \sup_{x \in \Omega} v_0(x)$ such that $\tilde{g}(t, x, 0, \rho_v) < 0$. It is easy to check that conditions (7.4) are satisfied for ρ_w, ρ_v defined above and

$$(K(x)u_0(x), v_0(x)) \in \mathbf{S}_\rho := \{(u_1, u_2) \in C([0, \infty) \times \overline{\Omega}); (0, 0) \leq (w, v) \leq (\rho_w, \rho_v)\}.$$

Therefore all the conditions of Theorem 16 in Appendix are satisfied, thus there exists a unique solution (w, v) of (2.7), and it is positive. Obviously, $(u, v) = (Kw, v)$ is the unique positive solution of (2.1) in $\mathbf{S}_\rho$. $\square$

Remark 1. *It is easy to check that if the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1), (A3) and (A5), they satisfy the assumptions of the theorem.*

3 Evolutionary Stability

In this section we will state the results on stability of the semi-trivial equilibria for system (2.1). Some preliminary results on the existence of equilibria are presented in Section 3.1.

Our approach to establishing an evolutionary advantage is to study the stability of so-called semi-trivial equilibria of the system (2.1), which are $(\tilde{u}, 0)$, $(0, \tilde{v})$, when only one species survives. It is easy to see that the functions $\tilde{u}$ and $\tilde{v}$ are the solutions of the following elliptic boundary value problems

$$\begin{cases} D\Delta \left(\frac{\tilde{u}(x)}{K(x)} \right) + \tilde{u}(x)g(x, \tilde{u}(x), 0) = 0, & x \in \Omega, \\ \frac{\partial(\tilde{u}/K)}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (3.1)$$

$$\begin{cases} \nabla \cdot d(x)\nabla \tilde{v}(x) + \tilde{v}(x)g(x, 0, \tilde{v}(x)) = 0, & x \in \Omega, \\ \frac{\partial \tilde{v}}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \quad (3.2)$$

and

$$\begin{cases} \nabla \cdot d(x)\nabla \tilde{v}(x) + \tilde{v}(x)g(x, 0, \tilde{v}(x)) = 0, & x \in \Omega, \\ \frac{\partial \tilde{v}}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \quad (3.3)$$

$$\begin{cases} \frac{\partial \tilde{v}}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \quad (3.4)$$

respectively.

It is also convenient for future analysis to introduce the substitution $\tilde{w} = \tilde{u}/K$. The steady-state problem (3.1), (3.2) for $\tilde{w}$ becomes

$$\begin{cases} \frac{D}{K(x)}\Delta \tilde{w}(x) + \tilde{w}(x)\tilde{g}(x, \tilde{w}(x), 0) = 0, & x \in \Omega, \\ \frac{\partial \tilde{w}}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (3.5)$$

$$\begin{cases} \frac{\partial \tilde{w}}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (3.6)$$

where we denote $\tilde{g}(x, w, v) = g(x, Kw, v)$.

3.1 Existence and uniqueness of equilibria

Lemma 1. *Let the growth function g satisfy the assumptions (A1)-(A5). The function $\tilde{u}(x) = K(x)$ is the only positive solution of (3.1)-(3.2).*

Proof. According to equations (3.1)-(3.2), the function $\tilde{u}$ is an equilibrium solution of scalar system (2.3)-(2.4). Therefore by Theorem 2 the unique positive solution of (3.1)-(3.2) is $\tilde{u}(x) = K(x)$. $\square$

Lemma 2. *Let the growth function g satisfy the assumptions (A1), and (A3)-(A5). There exists a unique positive solution $\tilde{v}(x) = v^*(x)$ of (3.3)-(3.4). Moreover, if $K(x) \not\equiv \text{const}$ then*

$$\int_{\Omega} g(x, 0, v^*)K(x)dx > 0 \quad (3.7)$$

Proof. Noting that the solution $\tilde{v}(x)$ of (3.3)-(3.4) is also an equilibrium solution of (2.6) and using Theorem 4, we obtain the first statement of the lemma. By (3.3) we have

$$\nabla \cdot d(x) \nabla v^* + v^* g(x, 0, v^*) = 0$$

Integrating over Ω and using (3.4) which implies $\int_{\Omega} \nabla \cdot d(x) \nabla v^* dx = 0$, we get

$$\int_{\Omega} g(x, 0, v^*) v^* dx = 0 \quad (3.8)$$

or applying the equality in (A3) we have

$$\int_{\Omega} G(x, v^*) v^* dx = 0$$

Integrating the equality

$$G(x, v^*) v^* = G(x, v^*) (v^* - K) + G(x, v^*) K$$

over Ω and using (3.8) we obtain

$$0 = \int_{\Omega} G(x, v^*) (v^* - K) dx + \int_{\Omega} G(x, v^*) K dx. \quad (3.9)$$

Using the Mean Value Theorem and the fact that $G(x, K) = 0$ by (A4) we can write

$$G(x, v^*) = G(x, v^*) - G(x, K) = G_v(x, \xi) (v^* - K) \quad (3.10)$$

where $\xi(x)$ is between $v^*(x)$ and $K(x)$ for each $(t, x) \in (0, \infty) \times \Omega$. Substituting (3.10) in (3.9) we therefore obtain

$$\int_{\Omega} G(x, v^*) K(x) dx = - \int_{\Omega} G_v(x, \xi) (v^* - K)^2 dx > 0$$

unless $v^*(x) \equiv K(x)$ where for the last inequality we used property (A5). One can check that $K(x)$ is not a solution of (3.3)-(3.4) unless $K = \text{const}$ which concludes the proof. $\square$

Remark 2. Note that by property (A5) it follows that $g(x, 0, v) \geq 0$ for any $v \leq K$. Therefore, one can think about condition (3.7) as the condition $v^* < K$ in an integral sense.

The next result is concerned with a coexistence state for system (2.1) of competing species.

Theorem 6. *Let the growth function g satisfy the assumptions (A1), and (A3)-(A5). If $K(x) \not\equiv \text{const}$ then there is no coexistence state (u_s, v_s) for system (2.1).*

Proof. Let us assume the contrary that there exists a strictly positive equilibrium solution (u_s, v_s) of (2.1). Then (u_s, v_s) satisfies

$$\begin{cases} D\Delta\left(\frac{u_s(x)}{K(x)}\right) + u_s(x)g(x, u_s(x), v_s(x)) = 0, & x \in \Omega, \\ \nabla \cdot d(x)\nabla v_s(x) + v_s(x)g(x, u_s(x), v_s(x)) = 0, & x \in \Omega, \\ \frac{\partial(u_s/K)}{\partial n} = d(x)\frac{\partial v_s}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (3.11)$$

By adding the first two equations in (3.11) and using property (A3) we obtain

$$D\Delta\left(\frac{u_s(x)}{K(x)}\right) + \nabla \cdot d(x)\nabla v_s(x) + (u_s(x) + v_s(x))G(x, u_s(x) + v_s(x)) = 0$$

Integrating over Ω and using the boundary conditions in (3.11) we obtain

$$\int_{\Omega} (u_s(x) + v_s(x))G(x, u_s(x) + v_s(x))dx = 0$$

Proceeding as in the proof of Lemma 2 with $u_s + v_s$ in place of v^* we can show that the strict inequality

$$\int_{\Omega} G(x, u_s + v_s)Kdx = - \int_{\Omega} G_v(x, \xi)(u_s + v_s - K)^2dx > 0 \quad (3.12)$$

is valid for some ξ between $K(x)$ and $u_s(x) + v_s(x)$ unless $u_s(x) + v_s(x) \equiv K(x)$. Therefore, we need to consider the following two cases.

1. If $u_s(x) + v_s(x) \equiv K(x)$, then $w_s = u_s/K$ and v_s satisfy

$$\begin{aligned} \Delta w_s &= 0, & x \in \Omega & \quad \nabla \cdot d(x)\nabla v_s = 0, & x \in \Omega \\ \frac{\partial w_s}{\partial n} &= 0, & x \in \partial\Omega & \quad \frac{\partial v_s}{\partial n} = 0, & x \in \partial\Omega \end{aligned}$$

and therefore by the Maximum Principle $w_s = \text{const}$, $v_s = \text{const}$, see e.g. Theorem 3.6 in the book of Gilbarg and Trudinger (1983). Since $w_s K + v_s = K$ and K is not constant, it follows that $v_s = 0$, $w_s = 1$, which contradicts $v_s > 0$.

2. Let $u_s(x) + v_s(x) \not\equiv K(x)$. Consider the eigenvalue problem

$$\begin{aligned} \frac{D}{K(x)}\Delta\psi(x) + \psi(x)\tilde{g}(x, w_s(x), v_s(x)) &= \sigma\psi(x), & x \in \Omega \\ \frac{\partial\psi}{\partial n} &= 0, & x \in \partial\Omega \end{aligned} \quad (3.13)$$

According to the variational characterization of eigenvalues described in Theorem 2.1 by Cantrell and Cosner (2003), its principal eigenvalue is given by

$$\sigma_1 = \sup_{\psi \neq 0, \psi \in W^{1,2}} \frac{-\int_{\Omega} D|\nabla \psi|^2 dx + \int_{\Omega} K \psi^2 \tilde{g}(x, w_s, v_s) dx}{\int_{\Omega} K \psi^2 dx}$$

Upon substituting $\psi = 1$ and using (3.12), we obtain

$$\sigma_1 \geq \frac{\int_{\Omega} K \tilde{g}(x, w_s, v_s) dx}{\int_{\Omega} K dx} > 0 \quad (3.14)$$

However, since (w_s, v_s) is an equilibrium solution of (2.7), w_s satisfies

$$\begin{aligned} \frac{D}{K(x)} \Delta w_s(x) + w_s \tilde{g}(x, w_s, v_s) &= 0, & x \in \Omega \\ \frac{\partial w_s}{\partial n} &= 0, & x \in \partial\Omega \end{aligned}$$

and is therefore a positive principal eigenfunction of (3.13) with the principal eigenvalue 0. This is a contradiction with (3.14). □

3.2 Stability analysis

In this section we describe all possible equilibrium states of system (2.1) and discuss their stability. We first state our main result.

Theorem 7. *Let the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1)-(A5). The semi-trivial equilibrium $(K, 0)$ of (2.1) is globally asymptotically stable. That is for any nonnegative nontrivial $u_0, v_0 \in C(\overline{\Omega})$ the solution $(u(t, x), v(t, x))$ of (2.1), (2.2) satisfies*

$$(u, v) \rightarrow (K, 0) \quad \text{as } t \rightarrow \infty$$

uniformly in $x \in \overline{\Omega}$.

The proof is postponed until the end of the section. The next theorem deals with the stability of the second semi-trivial equilibrium of (2.1). After the substitution $w(t, x) = u(t, x)/K(x)$, system (2.1) becomes

$$\begin{cases} \frac{\partial w(t, x)}{\partial t} = \frac{D}{K(x)} \Delta w(t, x) + w(t, x) \tilde{g}(x, w(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial v(t, x)}{\partial t} = \nabla \cdot d(x) \nabla v(t, x) + v(t, x) \tilde{g}(x, w(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial w}{\partial n} = d(x) \frac{\partial v}{\partial n} = 0, & x \in \partial\Omega. \end{cases} \quad (3.15)$$

where $\tilde{g}(x, w(t, x), v(t, x)) = g(x, K(x)u(t, x), v(t, x))$.

Theorem 8. *The semi-trivial equilibrium $(0, v^*(x))$ of (2.1) is unstable.*

Proof. Once again, we consider problem (3.15) obtained after the substitution $w = u/K$. The semi-trivial equilibrium of (3.15) corresponding to the semi-trivial equilibrium $(0, v^*(x))$ of (2.1) is again $(0, v^*(x))$. Since $K(x)$ is strictly positive and bounded from above in $\bar{\Omega}$, stability of the equilibrium $(0, v^*(x))$ of system (2.1) is equivalent to stability of the equilibrium $(0, v^*(x))$ of (3.15) and therefore we will investigate the latter system. To this end, let us consider the linearization of (3.15) around $(0, v^*(x))$

$$\frac{\partial w(t, x)}{\partial t} = \frac{D}{K(x)} \Delta w(t, x) + \tilde{g}(x, 0, v^*) w(t, x), \quad t > 0, \quad x \in \Omega, \quad (3.16)$$

$$\frac{\partial v(t, x)}{\partial t} = \nabla \cdot d(x) \nabla v(t, x) + \tilde{g}(x, 0, v^*) v(t, x) + v^*(x) \tilde{g}_v(x, 0, v^*) v(t, x) \quad (3.17)$$

$$+ v^*(x) \tilde{g}_w(x, 0, v^*) w(t, x), \quad t > 0, \quad x \in \Omega, \quad (3.18)$$

$$\frac{\partial w}{\partial n} = d(x) \frac{\partial v}{\partial n} = 0, \quad x \in \partial\Omega \quad (3.19)$$

and study the associated eigenvalue problem

$$\begin{aligned} \frac{D}{K(x)} \Delta \psi(x) + \tilde{g}(x, 0, v^*) \psi(x) &= \sigma \psi(x), \quad x \in \Omega \\ \frac{\partial \psi}{\partial n} &= 0, \quad x \in \partial\Omega \end{aligned} \quad (3.20)$$

$$\begin{aligned} \nabla \cdot d(x) \nabla \phi(x) + \tilde{g}(x, 0, v^*) \phi(x) + v^*(x) g_v(x, 0, v^*) \phi(x) \\ + v^*(x) g_w(x, 0, v^*) \psi(x) &= \sigma \phi(x), \quad x \in \Omega \\ d(x) \frac{\partial \phi}{\partial n} &= 0, \quad x \in \partial\Omega \end{aligned} \quad (3.21)$$

If the principal eigenvalue is positive then the semi-trivial equilibrium $(0, v^*)$ is unstable. Consider the first equation in (3.20), according to the variational characterization of the eigenvalues (Cantrell and Cosner 2003), the principal eigenvalue is given by

$$\sigma_1 = \sup_{\psi \neq 0, \psi \in W^{1,2}} \frac{- \int_{\Omega} D |\nabla \psi|^2 dx + \int_{\Omega} K \psi^2 \tilde{g}(x, 0, v^*) dx}{\int_{\Omega} K \psi^2 dx}.$$

Upon substituting $\psi = 1$ and using (3.7), we have $\sigma_1 \geq \frac{\int_{\Omega} \tilde{g}(x, 0, v^*) K dx}{\int_{\Omega} K dx} > 0$, which concludes the proof. $\square$

The following theorem deals with the trivial equilibrium.

Theorem 9. *The trivial equilibrium $(0, 0)$ of system (2.1) is unstable. Moreover, it is a repelling equilibrium.*

Proof. Similar to the proof of Theorem 8, we linearize system (3.15) around the origin to obtain the following eigenvalue problem:

$$\begin{aligned}
\frac{D}{K(x)} \Delta \psi(x) + \tilde{g}(x, 0, 0) \psi(x) &= \sigma \psi(x), \quad x \in \Omega \\
\frac{\partial \psi}{\partial n} &= 0, \quad x \in \partial \Omega \\
\nabla \cdot d(x) \nabla \phi(x) + \tilde{g}(x, 0, 0) \phi(x) &= \sigma \phi(x), \quad x \in \Omega \\
d(x) \frac{\partial \phi}{\partial n} &= 0, \quad x \in \partial \Omega
\end{aligned} \tag{3.22}$$

Integrating (3.22) we obtain

$$\sigma_1 = \frac{\int_{\Omega} \tilde{g}(x, 0, 0) \phi_1 dx}{\int_{\Omega} \phi_1 dx}$$

where ϕ_1 is the principal eigenfunction of (3.22), and ϕ_1 can be chosen positive. Therefore, $\sigma_1 > 0$ and the trivial equilibrium of (2.1) is unstable. Next, we want to show that the trivial equilibrium $(0, 0)$ is a repeller according to the definition (2) of Theorem 17. Let $\delta = \inf_{x \in \Omega} K(x)/4 > 0$ and let $u_0(x) \geq 0, v_0(x) \geq 0$ be such that $u_0(x) + v_0(x) < \delta$ and $(u_0(x), v_0(x)) \neq (0, 0)$. Let us add the first two equations of the system (2.1) and then integrate over the domain Ω using the Neumann boundary conditions. We obtain

$$\frac{d}{dt} \int_{\Omega} (u(t, x) + v(t, x)) dx = \int_{\Omega} (u(t, x) + v(t, x)) g(x, u(t, x), v(t, x)) dx$$

Next, we note that $\rho = \inf_{x \in \Omega} g(x, \delta, \delta) > 0$ and as long as $u(t, x) + v(t, x) \leq 2\delta$, we have

$$\frac{d}{dt} \int_{\Omega} (u(t, x) + v(t, x)) dx \geq \rho \int_{\Omega} (u(t, x) + v(t, x)) dx$$

Thus using Gronwall's lemma we obtain

$$\int_{\Omega} (u(t, x) + v(t, x)) dx \geq e^{\rho t} \int_{\Omega} (u_0(x) + v_0(x)) dx$$

And since $\int_{\Omega} (u_0(x) + v_0(x)) dx > 0$, we obtain that the integral $\int_{\Omega} (u(t, x) + v(t, x)) dx$ grows exponentially with time as long as $u(t, x) + v(t, x) \leq 2\delta$. We therefore conclude that there exists $t_0 > 0$ such that $u(t_0, x) + v(t_0, x) > \delta$ for some $x \in \Omega$ and hence the trivial equilibrium is a repeller. $\square$

We are now ready to prove the main result.

Proof of Theorem 7. Define the operator T_t by $T_t(\mathbf{u}_0) = \mathbf{u}$, where $\mathbf{u}_0 \equiv (u_0, v_0)$ and $\mathbf{u} \equiv (u, v)$ is a solution of (2.1)-(2.2). According to Lemma 5 in Appendix, the first

condition on T_t of Theorem 17 is satisfied. Next, condition (2) is guaranteed by Theorem 9, and condition (3) follows from Theorems 1-4. Finally, condition (4) can be deduced from Theorem 5 and Lemma 5 and therefore system (2.1) is a strongly monotone dynamical system and we can apply Theorem 17 from Appendix. Since the semi-trivial equilibrium $(0, v^*)$ is unstable and therefore is not asymptotically stable, it excludes the possibility of (c) in Theorem 17. Finally, there is no coexistence state therefore the equilibrium $(K, 0)$ is globally asymptotically stable. $\square$

4 Symmetric competition

In this section we consider a different system of competing populations, when either species might adopt a carrying capacity driven diffusion strategy. In this case we also show that the species adopting a carrying capacity driven diffusion strategy always survives in the competition, provided the second species does not adopt this strategy.

Namely, we consider

$$\begin{cases} \frac{\partial u(t, x)}{\partial t} = D_1 \Delta \left(\frac{u(t, x)}{P(x)} \right) + u(t, x)g(x, u(t, x), v(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial v(t, x)}{\partial t} = D_2 \Delta \left(\frac{v(t, x)}{Q(x)} \right) + v(t, x)g(x, v(t, x), u(t, x)), & t > 0, x \in \Omega, \\ \frac{\partial(u/P)}{\partial n} = \frac{\partial(v/Q)}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (4.1)$$

with the initial conditions

$$u(0, x) = u_0(x), \quad v(0, x) = v_0(x). \quad (4.2)$$

As before, we assume that $D_1 > 0$, $D_2 > 0$, $P(x)$ and $Q(x)$ are in the class $C^{1+\alpha}(\overline{\Omega})$, $P(x) > 0$, $Q(x) > 0$ for any $x \in \overline{\Omega}$. Here Ω is a an open nonempty bounded domain of $\mathbb{R}^n$ with $\partial\Omega \in C^{2+\alpha}$, $0 < \alpha < 1$. The functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ will satisfy some of the assumptions (A1)-(A5).

4.1 Existence and uniqueness

The first result is on the existence and uniqueness of solutions.

Theorem 10. *Let the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1), (A2), and, in addition, $g(x, u, v)$ be strictly decreasing in u and v . Then for any $u_0(x), v_0(x) \in C(\Omega)$, problem (4.1) has a unique solution (u, v) . Moreover, if the initial condition $(u_0(x), v_0(x))$ is nonnegative and nontrivial, then $u(t, x) > 0$, $v(t, x) > 0$ for any $t > 0$.*

Proof. Considering the substitution $w(t, x) = u(t, x)/P(x)$, $s(t, x) = v(t, x)/Q(x)$ we can proceed in the same way as in the proof of Theorem 5.

Choosing $\rho_w \geq \sup_{(t,x) \in Q_T} (u_0(t,x)/P(t,x))$ and $\rho_s > \sup_{x \in \Omega} (v_0(x)/Q(x))$ large enough and applying Theorem 16 from Appendix, we obtain a unique solution (w, s) of a substituted system in

$$\mathbf{S}_\rho := \{(w, s) \in C([0, \infty) \times \overline{\Omega}); (0, 0) \leq (w, s) \leq (\rho_w, \rho_s)\}.$$

Therefore, by the nondegeneracy of the substitution, there is a unique solution (u, v) to (4.1)-(4.2). $\square$

4.2 Equilibrium solutions

To obtain the main results about the equilibrium solutions we first need to state two preliminary lemmas.

Lemma 3. *Let the growth function g satisfy the assumptions (A1)-(A5), and let $P(x) \equiv K(x)$. The function $\tilde{u}(x) = K(x)$ is the only positive solution of*

$$\begin{cases} D\Delta \left(\frac{\tilde{u}(x)}{P(x)} \right) + \tilde{u}(x)g(x, \tilde{u}(x), 0) = 0, & x \in \Omega, \\ \frac{\partial(\tilde{u}/P)}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (4.3)$$

$$\begin{cases} \frac{\partial(\tilde{u}/P)}{\partial n} = 0, & x \in \partial\Omega \end{cases} \quad (4.4)$$

Lemma 4. *Let the growth function g satisfy the assumptions (A1), and (A3)-(A5). There exists a unique positive solution $\tilde{v}(x) = v^*(x)$ of*

$$\begin{cases} D\Delta \left(\frac{\tilde{v}(x)}{Q(x)} \right) + \tilde{v}(x)g(x, 0, \tilde{v}(x)) = 0, & x \in \Omega, \end{cases} \quad (4.5)$$

$$\begin{cases} \frac{\partial(\tilde{v}/Q)}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \quad (4.6)$$

Moreover, if $Q(x)/K(x) \not\equiv \text{const}$ then

$$\int_{\Omega} g(x, 0, v^*)K(x)dx > 0 \quad (4.7)$$

The proofs basically repeat the proofs of Lemmas 1 and 2, so we omit them here. We can now state the theorem about the coexistence state.

Theorem 11. *Let the growth function g satisfy the assumptions (A1), (A3)-(A5), and let $P(x) \equiv K(x)$, $\frac{Q(x)}{K(x)} \not\equiv \text{const}$. If $K(x) \not\equiv \text{const}$ then there is no coexistence state (u_c, v_c) for system (4.1).*

Proof. After the substitution $w(t, x) = u(t, x)/P(x)$, $s(t, x) = v(t, x)/Q(x)$ the proof proceeds in the same way as the proof of Theorem 6. We obtain that if there is a positive coexistence state (u_c, v_c) , only two cases are possible:

1. If $u_c(x) + v_c(x) \equiv K(x)$, then $w_c = u_c/K$ and $s_c = v_c/Q$ satisfy

$$\begin{aligned} \Delta w_c &= 0, & x \in \Omega & \qquad \Delta s_c = 0, & x \in \Omega \\ \frac{\partial w_c}{\partial n} &= 0, & x \in \partial\Omega & \qquad \frac{\partial s_c}{\partial n} = 0, & x \in \partial\Omega \end{aligned}$$

and therefore by the Maximum Principle $w_c = \text{const}$, $s_c = \text{const}$. Since $w_c K + s_c Q = K$, $Q/K \not\equiv \text{const}$ and K is not constant, it follows that $s_c = 0$, $w_c = 1$, which contradicts $s_c > 0$.

2. Let $u_c(x) + v_c(x) \not\equiv K(x)$.

Considering the eigenvalue problem

$$\begin{aligned} \frac{D}{K(x)} \Delta \psi(x) + \psi(x) \tilde{g}(x, w_c(x), s_c(x)) &= \sigma \psi(x), & x \in \Omega \\ \frac{\partial \psi}{\partial n} &= 0, & x \in \partial\Omega \end{aligned} \tag{4.8}$$

we can show that the principal eigenvalue σ_1 is strictly positive. This contradicts the fact that w_c is a positive principal eigenfunction of (4.8) with the zero principal eigenvalue.

□

The next two results deal with the semi-trivial equilibrium $(0, v^*)$ and the trivial equilibrium $(0, 0)$. The proofs are omitted since they repeat the proofs of Theorems 8 and 9.

Theorem 12. *Let all the conditions of Lemma 4 be satisfied. Then the semi-trivial equilibrium $(0, v^*(x))$ of (4.1) is unstable.*

Theorem 13. *Let the growth function g satisfy the assumptions (A1), and (A3)-(A5). Then the trivial equilibrium $(0, 0)$ of system (4.1) is unstable. Moreover, it is a repelling equilibrium.*

The main result of this section will now follow in the same way as Theorem 7.

Theorem 14. *Let the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1)-(A5), and let $P(x) \equiv K(x)$, $K(x) \not\equiv \text{const}$ and $Q(x)/K(x) \not\equiv \text{const}$. The semi-trivial equilibrium $(K, 0)$ of (4.1) is globally asymptotically stable. That is for any nonnegative nontrivial $u_0, v_0 \in C(\bar{\Omega})$ the solution $(u(t, x), v(t, x))$ of (4.1), (4.2) satisfies*

$$(u, v) \rightarrow (K, 0) \quad \text{as } t \rightarrow \infty$$

uniformly in $x \in \bar{\Omega}$.

Finally, to conclude the section we show that if $P/K \not\equiv \text{const}$ then there exists Q such the semi-trivial equilibrium $(u^*, 0)$ of (4.1) is unstable. This means that there are no local evolutionary stable strategies except those satisfying $P/K \equiv \text{const}$. This result thus complements Theorem 14 which says that $P = K$ is a globally evolutionary stable strategy.

Theorem 15. *Let the functions $f_1(x, u, v) = ug(x, u, v)$ and $f_2(x, u, v) = vg(x, v, u)$ satisfy (A1)-(A5), and let $P(x)/K(x) \not\equiv \text{const}$, $K(x) \not\equiv \text{const}$ and $Q(x) = K(x) + \varepsilon R(x)$ with $R(x) \in C^{1+\alpha}(\overline{\Omega})$ and $R(x) > 0$ in $\overline{\Omega}$. Then there exists $\varepsilon > 0$ such that the semi-trivial equilibrium $(u^*, 0)$ of (4.1) is unstable provided $R(x)$ satisfies*

$$\int_{\Omega} g(x, u^*, 0) R(x) dx > 0 \quad (4.9)$$

Proof. The proof is similar to the proof of Theorem 8 and we only present its sketch here. First note that if $P(x)/K(x) \not\equiv \text{const}$ then $(cK(x), 0)$ is not a solution to (4.1) for any constant $c > 0$. Therefore, we can assume that $u^*/K \not\equiv \text{const}$. Next, as in Lemma 2, we can show that

$$\int_{\Omega} g(x, u^*, 0) K(x) dx > 0.$$

Using condition (4.9) we can choose

$$\varepsilon = \frac{1}{2} \left| \frac{\int_{\Omega} g(x, u^*, 0) K(x) dx}{\int_{\Omega} g(x, u^*, 0) R(x) dx} \right|$$

to obtain

$$\int_{\Omega} g(x, u^*, 0) Q(x) dx = \int_{\Omega} g(x, u^*, 0) K(x) dx + \varepsilon \int_{\Omega} g(x, u^*, 0) R(x) dx > 0. \quad (4.10)$$

This condition can now be used to show that the principal eigenvalue of the linearization of (4.1) around $(u^*, 0)$ is positive. In the same way as in the proof of Theorem 8 we can show that the principal eigenvalue of the linearized problem satisfies

$$\sigma_1 \geq \frac{\int_{\Omega} g(x, u^*, 0) Q dx}{\int_{\Omega} Q dx} > 0,$$

which concludes the proof. □

5 Applications and Examples

5.1 Application to several growth laws

Some results of this paper in case of logistic growth function $G(x, u(t, x)) = r(x) \left(1 - \frac{u(t, x)}{K(x)}\right)$ were established by Korobenko et al. (2013). The intrinsic growth rate $r(x)$ and carrying capacity $K(x)$ are assumed to be positive Hölder continuous bounded functions. Below we list some of the other common growth functions, all of which as it is easy to check satisfy properties (A1)-(A5).

1. Gilpin-Ayala growth introduced by Gilpin and Ayala (1973)

$$G(x, u(t, x)) = r(x) \left(1 - \left(\frac{u(t, x)}{K(x)}\right)^\theta\right), \quad 0 < \theta < 1 \quad (5.1)$$

2. Nicholson's blowflies type model which is a nondelayed version of the equation considered by Nicholson (1954) and Gurney et al. (1980)

$$G(x, u(t, x)) = r(x)(e^{\alpha(1-u(t, x)/K(x))} - 1) \quad (5.2)$$

3. Food-limited growth developed by Smith (1963)

$$G(x, u(t, x)) = r(x) \frac{1 - u(t, x)/K(x)}{1 + \beta u(t, x)/K(x)}, \quad \beta > 0 \quad (5.3)$$

4. Gompertz growth introduced by Gompertz (1825)

$$G(x, u(t, x)) = r(x) \ln \left(\frac{K(x)}{u(t, x)}\right). \quad (5.4)$$

5.2 Examples

As one of fitness measures, average population density at an equilibrium can be considered. In the following series of examples we compare population density for random and carrying capacity driven dispersals. The first example illustrates the case when the carrying capacity driven diffusion leads to a higher average population level compared to random dispersal. The second example describes the situation when random dispersal leads to a higher average population level. Thus, the relation of average population densities for populations choosing various dispersal strategies is an open problem which requires some additional research.

Finally, in example 3 we consider time-dependent carrying capacity where random diffusion not only may lead to higher average population levels but also to the ideal free distribution.

Example 1. *The function*

$$u(x) = \cos x + A, \quad x \in [0, \pi] \quad (5.5)$$

is a stationary solution of the equation with a regular diffusion and Gilpin-Ayala growth function

$$\frac{\partial u(t, x)}{\partial t} = D\Delta u(t, x) + ru(t, x) \left(1 - \left(\frac{u(t, x)}{K(x)} \right)^\theta \right), \quad t > 0, \quad x \in [0, \pi], \quad (5.6)$$

where

$$K(x) = \frac{r^{\frac{1}{\theta}} (\cos x + A)^{1+\frac{1}{\theta}}}{(r(\cos x + A) - D \cos x)^{\frac{1}{\theta}}}. \quad (5.7)$$

Here $A > 1$, $0 < \theta < 1$ and $A > \frac{r+D}{r}$, thus $K(x)$ is everywhere positive. One can also check that the solution satisfies the Neumann boundary conditions $\frac{\partial u}{\partial x} = 0$ at $x = 0, \pi$.

The stationary solution of the relevant equation with the carrying capacity driven diffusion

$$\frac{\partial u(t, x)}{\partial t} = D_1 \Delta \left(\frac{u(t, x)}{K(x)} \right) + r(x)u(t, x) \left(1 - \left(\frac{u(t, x)}{K(x)} \right)^\theta \right), \quad t > 0, \quad x \in [0, \pi], \quad (5.8)$$

equals $K(x)$ defined in (5.7), here D_1 is an arbitrary positive number. First, note that

$$K^\theta u^{1-\theta} = \frac{r(\cos x + A)^2}{r(\cos x + A) - D \cos x}$$

and, as it was shown in Example 1 by Korobenko and Braverman (2012),

$$A\pi = \int_0^\pi (\cos x + A) dx = \int_0^\pi u(x) dx < \int_0^\pi \frac{r(\cos x + A)^2}{r(\cos x + A) - D \cos x} dx = \int_0^\pi K^\theta u^{1-\theta} dx \quad (5.9)$$

Using the Hölder inequality with exponents $p = 1/\theta$, $q = 1/(1 - \theta)$ we have

$$\int_0^\pi K^\theta u^{1-\theta} dx \leq \left(\int_0^\pi K dx \right)^\theta \left(\int_0^\pi u dx \right)^{1-\theta}. \quad (5.10)$$

From (5.9), (5.10) we obtain

$$\int_0^\pi u(x) dx < \int_0^\pi K(x) dx$$

which illustrates that the carrying capacity driven diffusion leads to higher average population levels for a chosen class of models.

Example 2. Consider the carrying capacity driven diffusion model with the logistic growth

$$\frac{\partial u(t, x)}{\partial t} = D_1 \Delta \left(\frac{u(t, x)}{K(x)} \right) + ru(t, x) (K(x) - u(t, x)), \quad t > 0, \quad x \in \Omega, \quad (5.11)$$

$$\frac{\partial \left(\frac{u}{K} \right)}{\partial n} = 0, \quad t > 0, \quad x \in \partial\Omega, \quad (5.12)$$

and the problem with the classical diffusion

$$\frac{\partial v(t, x)}{\partial t} = D_2 \Delta v(t, x) + rv(t, x) (K(x) - v(t, x)), \quad t > 0, \quad x \in \Omega, \quad (5.13)$$

$$\frac{\partial v}{\partial n} = 0, \quad t > 0, \quad x \in \partial\Omega, \quad (5.14)$$

The equilibrium solution of (5.11), (5.12) is $u^* = K$. Let v^* be the equilibrium solution of (5.13), (5.14). Substituting $v = v^*$, dividing (5.13) by v^* and integrating we obtain

$$0 = D_2 \int_{\Omega} \frac{|\nabla v^*(x)|^2}{v^*(x)^2} dx + r \int_{\Omega} K(x) dx - r \int_{\Omega} v^*(x) dx$$

or

$$\int_{\Omega} v^*(x) dx = \int_{\Omega} K(x) dx + \frac{D_2}{r} \int_{\Omega} \frac{|\nabla v^*(x)|^2}{v^*(x)^2} dx > \int_{\Omega} K(x) dx$$

provided $\nabla v^*(x) \not\equiv 0$. Thus, it is possible that the average equilibrium population level is greater for a randomly diffusing population compared to the population with a carrying capacity driven diffusion.

The next example is a generalization of Example 2 by Korobenko and Braverman (2012) to the case of the Gilpin-Ayala growth function. It shows that for time-dependent carrying capacities, a regular diffusion can lead to higher average population levels.

Example 3. Consider the carrying capacity driven diffusion model

$$\frac{\partial u(t, x)}{\partial t} = D_1 \Delta \left(\frac{u(t, x)}{K(t, x)} \right) + r(x)u(t, x) \left(1 - \left(\frac{u(t, x)}{K(t, x)} \right)^{\theta} \right), \quad t > 0, \quad x \in \Omega, \quad (5.15)$$

$$\frac{\partial \left(\frac{u}{K} \right)}{\partial n} = 0, \quad t > 0, \quad x \in \partial\Omega, \quad (5.16)$$

and the problem with the classical diffusion

$$\frac{\partial v(t, x)}{\partial t} = D \Delta v(t, x) + r(x)v(t, x) \left(1 - \left(\frac{v(t, x)}{K(t, x)} \right)^{\theta} \right), \quad t > 0, \quad x \in \Omega, \quad (5.17)$$

$$\frac{\partial v}{\partial n} = 0, \quad t > 0, \quad x \in \partial\Omega, \quad (5.18)$$

where $\Omega = [0, \pi]$, $r(x) \equiv r > 0$, $0 < \theta < 1$, and

$$K(t, x) = e^{-Dt} \cos(x) + A, \quad A > 2.$$

Define the average of any function $f(t, x)$ as

$$\langle f \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_{\Omega} f(t, x) dx dt$$

when such a limit exists. We will show that there are solutions $u(t, x)$ and $v(t, x)$ of (5.15)-(5.16) and (5.17)-(5.18), respectively, such that $v(0, x) = u(0, x)$ and the inequality $\langle u \rangle < \langle v \rangle$ holds. First note that $v(t, x) = K(t, x)$ is a solution of (5.17)-(5.18) with the initial condition $v(0, x) = \cos(x) + A$, and $K(t, x)$ is not a solution of (5.15)-(5.16). Then, integrating equation (5.15) from 0 to T in t and from 0 to π in x and using (5.16), we have

$$\int_0^{\pi} (u(T, x) - u(0, x)) dx = \int_0^T \int_0^{\pi} r u \left(1 - \left(\frac{u}{K} \right)^{\theta} \right) dx dt \quad (5.19)$$

Next, it has been shown in Theorem 1 by Korobenko et al. (2013) that there exists an upper solution of (5.15)-(5.16) and therefore $u(T, x)$ is uniformly bounded for any $T > 0$. Let $\delta = K^{\theta} u^{1-\theta} - u$, and note that we can write

$$\left(\frac{u}{K} \right)^{\theta} = 1 - \frac{\delta}{K^{\theta} u^{1-\theta}}$$

Next, divide the equation (5.19) by T and take the limit as $T \rightarrow \infty$ to obtain

$$\begin{aligned} 0 &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} r \left(\frac{u}{K} \right)^{\theta} (K^{\theta} u^{1-\theta} - u) dx dt, \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} r \delta \left(1 - \frac{\delta}{K^{\theta} u^{1-\theta}} \right) dx dt \end{aligned}$$

or

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} r \delta dx dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} r \frac{\delta^2}{K^{\theta} u^{1-\theta}} dx dt > 0,$$

since $\delta \not\equiv 0$. Therefore,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} K^{\theta} u^{1-\theta} dx dt > \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^{\pi} u dx dt \quad (5.20)$$

Using Hölder inequality with exponents $p = 1/\theta$, $q = 1/(1 - \theta)$ we have

$$\int_0^T \int_0^\pi K^\theta u^{1-\theta} dx dt \leq \left(\int_0^T \int_0^\pi K dx dt \right)^\theta \left(\int_0^T \int_0^\pi u dx dt \right)^{1-\theta}. \quad (5.21)$$

From (5.20), (5.21) we obtain

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^\pi u(t, x) dx dt < \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \int_0^\pi K(t, x) dx dt,$$

where $u(t, x)$ is a solution of (5.15)-(5.16) with any nontrivial initial condition, in particular, with $u(0, x) = \cos(x) + A = v(0, x)$. Hence, $\langle v \rangle = \langle K \rangle > \langle u \rangle$.

6 Discussion and Open Problems

Chen et al. (2012) studied the dynamics of a reaction-diffusion-advection model for two competing species in a spatially heterogeneous environment. Similarly to the present paper, the two species were assumed to have the same population dynamics but different dispersal strategies: both species disperse by random diffusion and advection along the environmental gradient, with different random dispersal and/or advection rates, and it was illustrated by Chen et al. (2012) that both coexistence and competitive exclusion are possible. We have shown that resources-directed dispersal is an evolutionarily stable strategy, independently of the actual value of the constant diffusion coefficients. This is the situation when all other parameters are the same for both types of species, in particular, intrinsic growth rates and time-dependent carrying capacities. Once growth rates are different and space-heterogeneous, a positive coexistence equilibrium may exist, which significantly changes the dynamics of the system. It would also be interesting to study the system with two different dispersal strategies, the same intrinsic growth rates and two different carrying capacities. The latter problem can be reformulated in the following way: what harvesting load should be imposed on the population with the carrying capacity driven diffusion to alleviate its evolutionary advantage and avoid invasion which would occur without harvesting?

Would the results of the present paper be the same if

- a) system of the form (2.1) includes time delays;
- b) both populations are harvested at the same rate;
- c) there are seasonal changes in the carrying capacity, i.e. $K(t, x)$ is T -periodic in t ;
- d) the growth rate g does not satisfy (A5) but can be negative for small u and v (systems with the Allee effect);

- e) the per capita growth function $g(x, u, v)$ is not symmetric in u, v , for example, $g(x, u, v) = G(x, \alpha u + \beta v)$?

A different approach to evolutionary stability of a given strategy is to consider a small perturbation to the strategy and ask if it is still stable under perturbation. This approach was used in Theorems 1,2 by Cantrell et al. (2010) and more recently by Averill et al. (2012). In the setting of (4.1) it would be interesting to consider the following question. Let $P = K + \alpha R$, and $Q = K + \beta R$, $R \not\equiv \text{const}$. Under what conditions on α, β

- a) will the equilibrium $(u^*, 0)$ be stable and $(0, v^*)$ be unstable?
b) will both $(u^*, 0)$ and $(0, v^*)$ be unstable and the system (4.1) have at least one stable positive steady state?

Let us note that the system considered by Cantrell et al. (2010) and Averill et al. (2012) is different from (4.1), however, the methods used by the authors can still be applied to our models and this would be an interesting question to consider. The method of Lyapunov functionals was recently employed to study n -species systems in Lou and Munther (2012). Model (4.1) of the present paper can be easily modified to the case of n species as well, where one of the species chooses K as a target, while the others are driven by some functions P_j which are not proportional to K . We suggest that this system can be studied by the method of Lou and Munther (2012) based on constructing a Lyapunov functional.

7 Appendix

In this section we present auxiliary results used in the proofs of Sections 2.2 and 3. The next theorem from the monograph by Pao (1992) deals with the time-dependent solution of the system of the form

$$\begin{aligned} \frac{\partial u_i}{\partial t} - L_i u_i &= f_i(x, u_1, u_2), \quad t > 0, \quad x \in \Omega \\ \frac{\partial u_i}{\partial n} &= 0, \quad x \in \partial\Omega \\ u_i(0, x) &= u_{i,0}(x), \quad x \in \Omega, \quad i = 1, 2, \end{aligned} \tag{7.1}$$

where for $i = 1, 2$ the operators L_i defined as

$$L_i u := \sum_{i,j=1}^n a_{ij}(t, x) \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i(t, x) \frac{\partial u}{\partial x_i} \tag{7.2}$$

are uniformly elliptic, namely, there exist positive numbers λ and Λ such that for every vector $\xi = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n$

$$\lambda |\xi|^2 \leq \sum_{i,j=1}^n a_{ij}(t, x) \xi_i \xi_j \leq \Lambda |\xi|^2, \quad (t, x) \in [0, T] \times \bar{\Omega}. \tag{7.3}$$

We assume that the coefficients of L are Hölder continuous in $[0, T) \times \Omega$, $\forall T > 0$ and the vector-function (f_1, f_2) is continuously differentiable and monotone nonincreasing in $\mathbb{R}^+ \times \mathbb{R}^+$.

For any $\rho = (\rho_1, \rho_2) \in \mathbb{R}_+^2$ define

$$\mathbf{S}_\rho \equiv \{(u_1, u_2) \in C([0, \infty) \times \overline{\Omega}); (0, 0) \leq (u_1, u_2) \leq (\rho_1, \rho_2)\}$$

The next result is Theorem 8.7.2 in the book of Pao (1992).

Theorem 16. *Let (f_1, f_2) be a quasimonotone nonincreasing Lipschitz function in $\mathbf{S}_\rho$ and let (f_1, f_2) satisfy*

$$\begin{aligned} f_1(t, x, \rho_1, 0) &\leq 0 \leq f_1(t, x, 0, \rho_2) \\ f_2(t, x, 0, \rho_2) &\leq 0 \leq f_2(t, x, \rho_1, 0) \end{aligned} \tag{7.4}$$

for any $x \in \Omega$, $t > 0$. Then for any $(u_{1,0}, u_{2,0}) \in \mathbf{S}_\rho$ there exists a unique solution of (7.1) $\mathbf{u} \equiv (u_1, u_2)$ in $\mathbf{S}_\rho$, and $u_i(t, x) > 0$ for $x \in \Omega$, $t > 0$ when $u_{i,0} \neq 0$, $i = 1, 2$.

The next lemma (Cantrell and Cosner 2003) implies the monotonicity property for solutions of (7.1).

Lemma 5. *Let $(u_i(t, x), v_i(t, x))$, $i = 1, 2$, be two solutions of system (7.1) such that $u_1(0, x) \geq u_2(0, x)$ and $v_1(0, x) \leq v_2(0, x)$ for any $x \in \Omega$. Then $u_1(t, x) \geq u_2(t, x)$ and $v_1(t, x) \leq v_2(t, x)$ for any $x \in \Omega$ and any $t > 0$. Moreover, if*

$$(u_1(0, x), v_1(0, x)) \neq (u_2(0, x), v_2(0, x)),$$

then $u_1(t, x) > u_2(t, x)$ and $v_1(t, x) < v_2(t, x)$ for any $x \in \Omega$ and any $t > 0$

For the proof see Theorem 1.20 in the book by Cantrell and Cosner (2003) and remarks thereafter. The final statement of the lemma follows from the strong maximum principle for parabolic equations (Protter and Weinberger 1967) applied to the differences $u_1 - u_2$ and $v_2 - v_1$.

The next result by Hsu et al. (1996) classifies all possible equilibria for a monotone dynamical system. We will present a particular case of Theorem B (Hsu et al. 1996). Denote $X^+ = C_+(\overline{\Omega}) \times C_+(\overline{\Omega})$, where $C_+(\overline{\Omega})$ is the class of all nonnegative functions from $C(\overline{\Omega})$; $I = \langle 0, \tilde{u}_1 \rangle \times \langle 0, \tilde{u}_2 \rangle$, where $(\tilde{u}_1, 0)$ and $(0, \tilde{u}_2)$ are the semi-trivial equilibria of (7.1). Here $\langle \cdot, \cdot \rangle$ as in the proof of Theorem 5.

Theorem 17. *Let operator T_t be defined as $T_t(\mathbf{u}_0) = \mathbf{u}$, where $\mathbf{u}_0 \equiv (u_{1,0}, u_{2,0})$ and $\mathbf{u} \equiv (u_1, u_2) = (T_t(u_{1,0}), T_t(u_{2,0}))$ is a solution of (7.1). Let the following conditions hold:*

- (1) *T is strictly order preserving, which means that $u_1(x) \geq v_1(x)$ and $u_2(x) \leq v_2(x)$ imply $T_t(u_1(x)) \geq T_t(v_1(x))$ and $T_t(u_2(x)) \leq T_t(v_2(x))$.*

- (2) $T_t(\mathbf{0}) = \mathbf{0}$ for all $t \geq 0$ and $\mathbf{0}$ is a repelling equilibrium. That is there exists a neighbourhood U of $\mathbf{0}$ in X^+ such that for each $(u_1, u_2) \in U$, $(u_1, u_2) \neq \mathbf{0}$, there is $t_0 > 0$ such that $T_{t_0}(u_1, u_2) \notin U$.
- (3) $T_t((u_1, 0)) = (T_t(u_1), 0)$ and $T_t(u_1) \geq 0$ if $u_1 \geq 0$. There exists $\tilde{u}_1 > 0$ such that $T_t((\tilde{u}_1, 0)) = (\tilde{u}_1, 0)$ for any $t \geq 0$. The symmetric conditions hold for $T_t((0, u_2))$.
- (4) If $u_{i,0} \geq 0$, $u_{i,0} \neq 0$, $i = 1, 2$ then $T_t(u_{i,0}) > 0$, $i = 1, 2$. If $u_1(x) \geq v_1(x)$, $u_2(x) \leq v_2(x)$ and $u_1(x) \neq v_1(x)$, $u_2(x) \neq v_2(x)$ then $T_t(u_1(x)) > T_t(v_1(x))$ and $T_t(u_2(x)) < T_t(v_2(x))$.

Then exactly one of the following holds:

- (a) There exists a positive coexistence equilibrium $(u_{1,s}, u_{2,s})$ of (7.1).
- (b) $(u_1, u_2) \rightarrow (\tilde{u}_1, 0)$ as $t \rightarrow \infty$ for every $(u_{1,0}, u_{2,0}) \in I$.
- (c) $(u_1, u_2) \rightarrow (0, \tilde{u}_2)$ as $t \rightarrow \infty$ for every $(u_{1,0}, u_{2,0}) \in I$.

Moreover, if (b) or (c) holds then for every $(u_{1,0}, u_{2,0}) \in X^+ \setminus I$ and $u_{i,0} \neq 0$, $i = 1, 2$ either $(u_1, u_2) \rightarrow (\tilde{u}_1, 0)$ or $(u_1, u_2) \rightarrow (0, \tilde{u}_2)$ as $t \rightarrow \infty$.

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